ECA02 – DIGITAL CIRCUITS

CO4 PERFORMANCE EVALUATION ASSIGNMENT

TTL, CMOS, Memory Elements, PLDs and FPGA Architecture

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Course	ECA02 – Digital Circuits
Assignment	CO4 – Performance Evaluation

Prepared as a soft-copy assignment

Introduction

A digital system is built using logic gates, memory elements and programmable devices. Each part must receive valid voltage levels, operate within timing limits and consume acceptable power. This assignment evaluates a mixed system containing 74LS TTL logic, a 3.3 V CMOS controller, latches/flip-flops and programmable logic.

Important design note: The assignment does not identify exact IC part numbers or datasheets. Therefore, equations and architecture-level comparisons are provided. Numerical limits such as exact fan-out, propagation delay, input voltage tolerance and standby power must be confirmed from the datasheet of the selected devices.

Part (a) – TTL NAND Gate: Noise Margins, Fan-out and Temperature

1. Noise margin meaning

Noise margin is the amount of unwanted electrical noise a digital signal can tolerate without being read incorrectly. A TTL gate has guaranteed output voltage levels and required input voltage levels.

Symbol	Meaning
VOH(min)	Minimum guaranteed HIGH output voltage
VOL(max)	Maximum guaranteed LOW output voltage
VIH(min)	Minimum input voltage recognised as HIGH
VIL(max)	Maximum input voltage recognised as LOW

2. Noise-margin equations

HIGH noise margin: NMH = VOH(min) – VIH(min)

LOW noise margin: NML = VIL(max) – VOL(max)

A larger positive noise margin means the circuit can tolerate more noise. For a common 74LS-TTL example, typical guaranteed levels are VOH(min)=2.7 V, VOL(max)=0.5 V, VIH(min)=2.0 V and VIL(max)=0.8 V. Thus:

NMH = 2.7 – 2.0 = **0.7 V**; NML = 0.8 – 0.5 = **0.3 V**.

These are representative 74LS logic-level values; use the exact datasheet values for the selected gate.

3. Fan-out equation

Fan-out is the maximum number of similar logic inputs that one gate output can drive while maintaining valid logic levels.

Fan-out HIGH = |IOH(max)| / IIH(max)

Fan-out LOW = IOL(max) / |IIL(max)|

Allowed fan-out = the smaller of the HIGH and LOW fan-out values (rounded down to a whole number).

For example, if a gate can source 0.4 mA and each load needs 20 µA, the HIGH fan-out is 0.4 mA / 0.02 mA = 20. If it can sink 8 mA and each load draws 0.4 mA, LOW fan-out is 8/0.4 = 20. The theoretical fan-out is 20, provided timing and wiring capacitance are also acceptable.

4. Effect of higher temperature

• Leakage currents in semiconductor devices generally increase as temperature rises. This can increase the load current and reduce the number of inputs that one output can safely drive.

• Transistor characteristics and switching thresholds vary with temperature. Guaranteed output levels and noise margins must therefore be checked over the full operating-temperature range.

• Carrier mobility generally decreases as temperature rises, which can make internal switching slower and increase propagation delay.

• At high switching frequency, increased propagation delay and load capacitance can cause transitions to overlap or arrive too late. Dynamic power also increases with switching activity and capacitance.

Conclusion: Higher temperature can reduce practical fan-out and increase propagation delay. The safe design uses datasheet worst-case limits, adequate noise margin and a suitable thermal operating range.

Part (b) – Tristate TTL Bus Connected to a 3.3 V CMOS Controller

1. Tristate bus meaning

A tristate gate has three output conditions: logic 0, logic 1 and high impedance (Hi-Z). Hi-Z behaves like a disconnected output, allowing another device to use the shared bus.

2. Main electrical risks

• **Bus contention:** if two enabled drivers put opposite values on the bus at the same time, one tries to drive HIGH while the other drives LOW. This can cause excessive current, heating and incorrect data.

• **Voltage mismatch:** a 5 V TTL output may exceed the permitted input voltage of a 3.3 V CMOS pin. Even if the logic HIGH is recognised, the pin may not be 5 V tolerant.

• **Latch-up or input stress:** overvoltage or current injected into protection structures can cause abnormal current paths or damage, depending on the receiver design.

• **Signal integrity:** long bus wiring and high input capacitance can cause slow edges, ringing and timing errors.

3. Recommended interface

Use a dedicated mixed-voltage bus transceiver or level-shifting buffer that is explicitly specified for 5 V TTL-side signals and 3.3 V CMOS-side signals. Select a device with suitable input tolerance, output voltage levels, bus direction control and output-enable control.

• Connect the TTL bus to the transceiver's 5 V-tolerant input side.

• Connect the transceiver's 3.3 V side to the CMOS controller.

• Use the output-enable (OE) and direction controls so only one device drives the bus at a time.

• Use break-before-make control: disable the current driver before enabling the next driver.

• Check the transceiver datasheet for power-off isolation, maximum input voltage, output current and propagation delay.

Important: A simple series resistor or pull-up alone is not a complete substitute for a correctly specified level shifter. A resistor may limit current in some designs, but it does not guarantee safe logic levels or prevent all contention cases.

Conclusion: A correctly selected level-shifting bus transceiver with controlled output-enable signals is the reliable solution for the mixed-voltage bus.

Part (c) – Clock Skew, Latches, Flip-flops and Timing Violations

1. Clock skew

Clock skew is the difference in the arrival time of the same clock signal at two memory elements. It occurs because clock wires and internal circuits have different delays.

2. Level-sensitive latches versus edge-triggered flip-flops

Feature	Level-sensitive latch	Edge-triggered flip-flop
When it responds	While enable/clock is at its active level	Only near the active clock edge
Transparency	Input can pass to output during active level	Input is sampled at the triggering edge

Feature	Level-sensitive latch	Edge-triggered flip-flop
Skew sensitivity	A long transparent window can allow timing changes to pass through multiple stages	Mainly concerned with arrival differences between launch and capture edges
Timing concern	Transparency overlap and race-through	Setup and hold around the active edge

Latches can be useful for time borrowing, but if two neighbouring latches are transparent at the same time, data may pass through more than one stage. Flip-flops provide clearer stage boundaries, although clock skew still changes the available setup and hold margins.

3. Setup time and hold time

Setup time (tsu): minimum time the input data must remain stable before the active clock edge.

Hold time (th): minimum time the input data must remain stable after the active clock edge.

Violation	What happens	Possible result
Setup violation	Data changes too close to or after the sampling edge	Wrong value may be captured; metastability may occur
Hold violation	Data changes too soon after the sampling edge	Old data may not be captured reliably; metastability may occur

Metastability is a temporary uncertain output condition in which a memory element does not settle immediately to a valid 0 or 1. It can delay downstream logic or cause an incorrect state to be observed.

4. Effect of clock skew

• Positive skew (capture clock arrives later) can help setup timing but can make hold timing more difficult.

• Negative skew (capture clock arrives earlier) can make setup timing more difficult but may help hold timing.

• Designers use clock-tree balancing, timing analysis, suitable placement and synchronizer circuits for asynchronous inputs.

Conclusion: Both latches and flip-flops require correct timing. Latches need careful control of transparency windows; flip-flops need setup and hold requirements met at each active edge.

Part (d) – PLA versus PAL for Shared Product Terms

1. Basic structure

A Programmable Logic Array (PLA) and Programmable Array Logic (PAL) are programmable logic devices used to implement Boolean functions in sum-of-products form.

Device	AND array	OR array
PLA	Programmable	Programmable
PAL	Programmable	Fixed / pre-connected

2. Shared product terms

A product term is an AND combination of input variables or their complements. If two or more output functions use the same product term, that term may be generated once and shared.

Example: F1 = AB + AC and F2 = AB + AD. The term AB is common to both outputs. A PLA can create AB once in its programmable AND array and connect it to both output OR gates. This avoids generating the same term separately.

3. Comparison

Factor	PLA	PAL
Product-term sharing	Flexible; a common term can feed multiple outputs	More restricted by the fixed OR structure
Resource use for shared logic	Often efficient when many outputs share terms	May duplicate terms or run into per-output term limits
Layout / structure	More flexible but can require more programmable connections	More regular and often simpler

Factor	PLA	PAL
Speed	Extra programmable connections can add delay	Often faster due to simpler fixed OR connections
Best fit	Complex logic with heavily shared product terms	Straightforward logic with predictable per-output terms

Conclusion: For the stated algorithm with heavily shared product terms, a PLA is generally the better fit because its programmable AND and OR arrays allow sharing across outputs. A PAL may be preferable when the logic is simpler and speed or regular structure is more important. Actual resource and timing results depend on the specific device.

Part (e) – CPLD versus FPGA

1. Architecture

A CPLD is built from programmable logic blocks or macrocells connected by a relatively structured interconnect. An FPGA contains many configurable logic blocks (often LUT-based), registers and a larger programmable routing network.

2. Comparison

Parameter	CPLD	FPGA
Routing predictability	More structured and often more predictable; fewer routing choices	Large flexible routing mesh; delay depends more on placement and routing
Configuration memory	Commonly non-volatile Flash or EEPROM	Many mainstream devices use SRAM configuration
Power-up / boot	Often becomes operational quickly after power is applied	SRAM FPGA usually loads configuration from external non-volatile memory be
Standby static power	Often low for small CPLDs, but device-dependent	Can be higher due to larger fabric and configuration resources; device-depend
Typical use	Glue logic, control, simple state machines, instant-on functions	Large digital systems, DSP, parallel processing and complex datapaths

3. Configuration detail

A Flash/EEPROM-based CPLD retains its configuration without an external configuration load at every power-up. An SRAM-based FPGA generally loses its configuration when power is removed and must be configured again at startup. Some modern FPGAs also include non-volatile configuration options, so the exact part number matters.

Conclusion: Choose a CPLD when instant-on behavior and predictable control routing are important. Choose an FPGA when the design needs a large amount of logic, parallel processing or flexible datapaths. Compare actual standby-current and startup specifications before selection.

Part (f) – Mapping a 4-Variable Function into a 4-Input FPGA LUT

1. What is a LUT?

A Look-Up Table (LUT) is a small memory block inside an FPGA logic element. It stores the output value for every possible combination of its input variables. The LUT output therefore implements a Boolean function.

2. Number of SRAM bits

For n input variables, the LUT requires 2^n stored output bits. For four inputs:

Number of entries = 2^4 = 16; SRAM bits consumed = 16 × 1 = 16 bits.

3. Chronological CAD mapping steps

Step	CAD action
1. Enter function	Designer enters the Boolean expression or truth table for variables A, B, C and D.
2. Logic synthesis	CAD software simplifies the logic and converts it into a form suitable for FPGA resources.
3. LUT mapping	The mapper identifies that the function can fit into one 4-input LUT.
4. Truth-table generation	The tool evaluates the function for all 16 combinations of A, B, C and D.
5. Configuration-bit creation	Each truth-table output value becomes one configuration bit, giving 16 bits total.

Step	CAD action
6. Placement	The tool selects a physical configurable logic block for the LUT.
7. Routing	Programmable connections link the four input signals to the LUT inputs and connect its output to the next circuit.
8. Bitstream generation	The CAD tool creates the configuration data for the FPGA.
9. Programming	The configuration data is loaded into the FPGA's configuration SRAM. The LUT now produces the required output.

Conclusion: A generic 4-input LUT consumes exactly 16 configuration SRAM bits for its truth table. Additional FPGA routing and configuration resources are separate from these 16 LUT truth-table bits.

Part (g) – Unified Figure of Merit and Design Trade-offs

1. Why a Figure of Merit is needed

A digital design must balance speed, power, fan-in, noise margin, area and reliability. Improving one parameter can make another worse. For example, adding more loads can increase capacitance and delay; using a larger or faster device can increase power.

2. Example normalized Figure of Merit

A useful comparison framework is:

FoM = $(F/Fref)^{\alpha} \div [(D/Dref)^{\beta} \times (P/Pref)^{\gamma}]$

Where F is the supported fan-in or fan-out capability selected for comparison, D is propagation delay, P is power, and Fref, Dref and Pref are reference values. α, β and γ are weighting factors chosen according to the application. A higher FoM indicates a better balance under the chosen weights.

All quantities are normalized by reference values, so the ratio is dimensionless. The formula is a proposed engineering comparison metric, not a universal industry standard. It must be used together with hard electrical constraints.

3. Design constraints and trade-offs

Goal	Design consideration
High fan-in / fan-out	Use buffering or a suitable logic family; do not exceed input-current or capacitive-load limits.
Low propagation delay	Choose fast devices, reduce unnecessary logic levels and keep routing short; check worst-case timing.
Low power	Reduce switching activity and capacitance; select low-power devices and avoid unnecessary clocking.
Noise immunity	Maintain positive HIGH and LOW noise margins and use correct voltage translation.
Reliability	Meet setup/hold, temperature, voltage and current limits; prevent bus contention and metastability propagation.

4. Recommended mixed-technology design approach

• Keep the 74LS TTL subsystem within its specified supply, loading and temperature limits.

• Use a dedicated level-shifting transceiver between 5 V TTL signals and 3.3 V CMOS pins when direct compatibility is not guaranteed.

• Use one controlled bus driver at a time to prevent contention.

• Use edge-triggered registers or carefully designed latch stages with static timing analysis for synchronous data storage.

• Use a PLA when shared product terms are important; use a PAL for simpler logic where its fixed structure is sufficient.

• Use a CPLD for smaller instant-on control logic, or an FPGA for larger and more complex programmable functions.

• Verify timing, power and electrical margins with datasheets and CAD reports rather than relying only on typical values.

Overall conclusion: A reliable mixed TTL/CMOS/FPGA system is not designed by maximizing speed alone. It must satisfy voltage compatibility, noise margin, loading, timing, power and safety requirements together. The best architecture is the one that meets all required limits with adequate margin.

Final Conclusion

The assignment evaluated TTL noise margins and temperature effects, mixed-voltage tristate bus risks, clock skew and timing violations, PLA/PAL structures, CPLD/FPGA architectures and 4-input LUT mapping. The analysis shows that proper level shifting, controlled bus access, timing verification and suitable programmable-device selection are essential for a dependable digital system. Exact numerical performance must be checked against the datasheets and implementation reports of the selected components.

References / Further Reading

• ECA02 Digital Circuits – CO4 Performance Evaluation Assignment sheet supplied with this task.

• Manufacturer datasheets for the selected 74LS TTL gates, 3.3 V CMOS controller, bus transceiver, CPLD and FPGA.

• FPGA/CPLD CAD tool documentation for logic synthesis, LUT mapping, placement, routing and bitstream generation.

• Digital logic design textbook or course notes covering TTL/CMOS logic levels, timing, PLDs and FPGA architecture.

Submission note: This document answers all seven parts using the terminology and scope of the assignment. Add device-specific datasheet references and actual simulation/CAD reports if your instructor requires measured or tool-generated results.